*Using Capitec application to transfer money, the system instantly checks the database to verify account balance, another individual account balance and update both accounts.

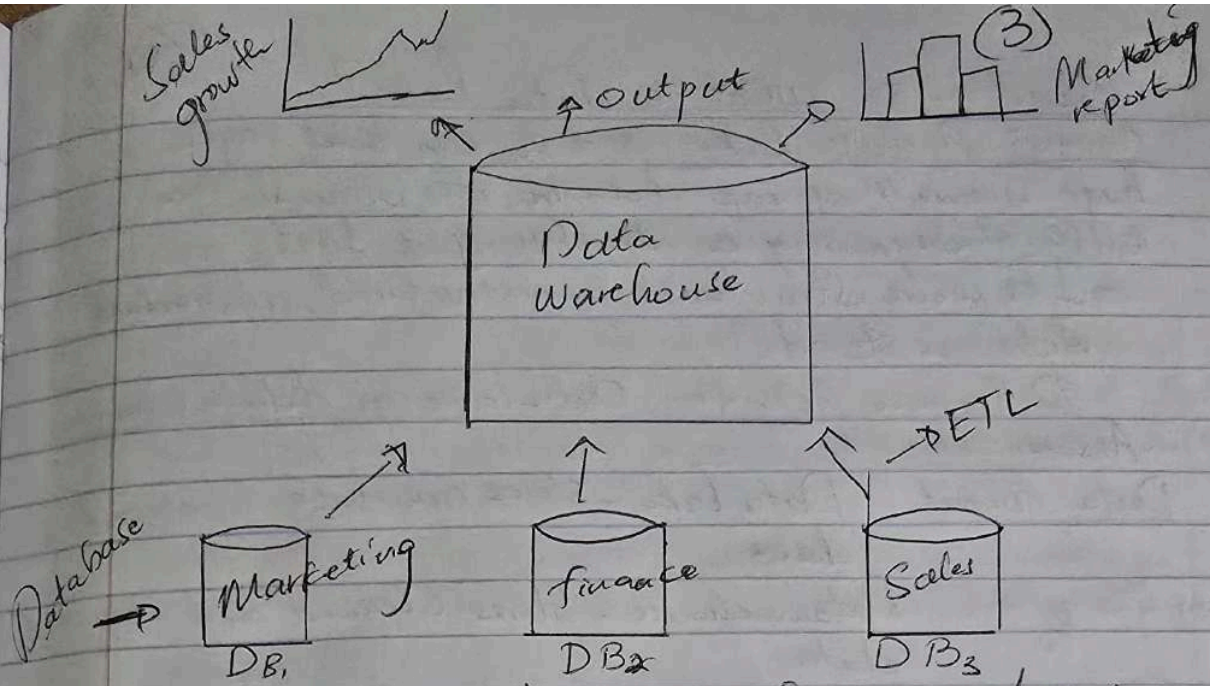
(2). Educational Institution: Schools uses the database to manage student records, Store student records, grades, attendence, who owes and who paid, fees, registrations, applications of incoming students and exclusions.
- BrightLearn used database to store Studet information, who paid R550, who registered for data science, data analytics and other Courses.

Question 2.

What is a data warehouse. How does it differ from regular database. Why would a Company use it.

What is a data warehouse:

Answer: Datawarehouse is a Central place where we stor multiple databases.
According to IBM, data warehouse aggregates data from various Sources into a Central data store optimized for querying and analyses.
* It typically uses extract, transform and load (ETL) or Extract, Load, and transform (ELT) processing to Clean, prepar, and organise data for business intelligence and data analytics use.



What is the difference between database and data warehouses? Answer.

- Database is used to store current, everyday data and is used to run daily operations.
  example - When Khutso transfers money to Lerato, via capitec APP, the system uses database.
- Data warehouse - is used to store large amounts of past historical data (from different databases) and is used to analyse and make decisions.
  Example - Khutso uses SQL to analyse data then use Power BI to visualize the analytics.

Why would a company use data warehouse?

Answer:

- For better decision-making - managers can analyse trends and patterns over time.
- analyse large amounts of data
- Easily create reports and dashboards for insights
- Identify trends and patterns.
- Combine data from different departments.

Question 3: What is data lake?

Answer: A data lake is a system that stores huge amounts of raw data in its original form without organising or structuring it first.
- Its where Structured, semistructured, unstructured data is stored.

Difference between Data lake and data warehouse

Answer:

Data format: Data lake - stores raw data. unprocessed data.
Datawarehouse - Stores cleaned and structured data.

Structure: Data-lake - flexible data structure.
Datawarehouse - Highly structured data

Processing: Data lake store first, analyse later
Data Warehouse - Clean first, then store and analyse.

When would the business Prefer data lake?

Answer:
- Business wants flexibility.
- Have massive data and diverse data.
- Need advanced analytics or Machine learning.

Question 4. What is a Data Mart

Answer.

Data Mart - is a smaller, focused version of a data warehouse that is designed for a specific department or business area.
Data Mart are used by specific groups within a company such as Sales team, finance team

(5)

How does it relates to a data warehouse.

Answer:

The data warehouse stores all the data in the company and data mart takes only the relevant part of the data for a specific department.

e.g. data warehouse is central storage and data mart acts as a smaller focused section built from that source.

~~- Marketing team outside Datawarehouse.~~

Marketing data mart for the marketing team.

Who typically uses it:

Answer:

Sales team - to track sales performance and targets

Marketing team - to analyse campaigns and customer behavior.

Finance team - to monitor budgets, costs and revenue.

Question 5 Explain difference between Structured, Semi-structured, and unstructured data.

Answer:

Structured data is highly organised and stored in a fixed format, usually in tables with rows and columns.

e.g. tests data. Structured data

Date	test-no.	Marks
mar 26	1	68%
Apr 26	2	78%
may 26	3	28%

Semi Structured data - is a partially organised data
- does not follow the format of a tabular model or relational databases because it does not have fixed Schema.
example - emails, JSON

Unstructured data - has no pre-defined format or organization.
- data is harder to store and analyse directly
e.g. Videos, images and audio files.

Type	Structured level	Example
Structured	Highly	Bank table
Semi-Structured	Partially	Emails
Unstructured	No Structure	Videos

Question 6:
What is data modelling?
Answer
Data modelling is the process of analyzing and defining all different types your business collects and produces, as well as the relationships between those bits of data.
- Creating a blueprint for how data will be structured before building a database or System

Why is data modeling important
- Clarity and organization
- prevents duplication and inconsistency in data
- Well Structured data makes Systems faster and easier to use.

Save time and cost.

Two types of data models.

(1) Conceptual data model - focuses on what data is needed, not how its stored.
- High level and simple.

(2) Logical data model - more details than conceptual
- Defines attributes, relationship and rules.

(3) Physical data model - Very detailed and technical.
- Shows how data is actually stored in a database
- Includes tables, columns, data types and indexes

Question 7.
What is data mining?

Answer:

Data mining is the process of analyzing large amounts of data to discover patterns, trends, and useful information that are not immediatly obvious.
↳ It mean digging through data to find hidden insights that help make better decision.

How is it used in business?

Data mining used by companies to;
- Understand customer behaviour (Buy, like or avoid)
- Improve marketing (targeting the right customers with right ads)
- Increase Sales (Recommend Products to Customers)
- Detect fraud (Identify unusual transactions)
- Improve decision making using real data.

Two real world examples

Answer.

(1). Online Shopping recommendations
↳ Shein recommends products to customers.
↳ Amazon uses data mining to analyse customer needs.

(2). Streaming Service recommendation.
↳ Netflix uses data mining
↳

(3) Social media recommendations.
↳ Instagram or Ticktok suggestion through data mining.

Question 8

What is a DMBS (Database Manggemntt Systems)

Answer.

*DMBS - is a Software that allows users to create, store, manage, and retrieve data from database efficiently and Securely.

DMBS helps with - Storing data, retrieving data, updating and deleting data, controlling access & Security and preventing data duplication and errors

Four popular DMBS Platforms and their uses.

Answer.

(1). MySQL

Description - Stores data in table (rows/columns) and uses SQL.

use case; Web application like online stores.
Example: Amazon using MySQL to manage users, products, and orders.

(2). PostreSQL - Advanced open source relation DMBS known for strong performance and flexibility.
Description: supports complex queries, large dataset, and advanced features like JSON support.
use case: Applications needing complex data analysis and financial systems.
Example: A banking system tracking transactions and account details with high accuracy.

(3) Oracle Database - powerful Commercial retional DBms used in large enterprises.
Description: Highly secure, scalable and designed for mission critical system.
Use case: Large corporations and government system
Example: A telecom company managing millions of customers records and billing system.

(4) - MongoDB - popular NOSQL DMBs that stores data in flexible document format.
Description: Does not use tables; store data docs making it highly flexible.
Use case; Big data, real time apps, and content heavy applications.
Example: A social media app like Meta able to store posts, comments and activity logs.

Question 9: Transform
What is ETL (Extract, Load)

Answer:

* ETL is a process used to collect data from different sources, clean and organise it and then store it in a system (like data warehouse) for analysis.

Three steps.

(1) * E (Extract) ; involves collecting data from various sources. Sources can include databases, Spreadsheets, websites or Apps.

(2) T (Transform) - This is where the data is cleaned and prepared.
- Removes errors, convert data into a consistent format, organise it and structure it, and apply rules.

(3) L (Load) - this is where cleaned data is ~~stored~~ Stored in a final system, usually a datawarehouse.
- Data is now ready for analysis and reporting.

Why ETL is important.
- Ensures data quality.
- Combines data from multiple Sources.
- Automates data preparation instead of doing it manually.
- Enables better analysis.
- Supports decision-making.

Question 10. (11)

(10) What are common data storage formats used for in analytics.

Answer.

(1) CSV (comma Separated Values).

- A plain text file where data is stored in rows and columns, separated by commas.

When to use: Simple data storage, and small datasets.

(2) JSON (Javascript Object Notation).

- A text format that stores data in a structured key value way.

When to use: Data with nested or complex structure. Web application and API

(3) Parquet.

- A Column based storage format optimized for big data processing.

When to use: Big data analytics, and when you need fast queries and high performance.

(4) Avro.

- A binary format that stores data along with its schema.

When to use: Data Streaming and data exchange between systems.

- Systems where data structure may change over time.

SECTION B: AI and Machine Learning.

Question 11
What is AI - Artifical Intelligence.

Answer:
AI or Artificial intelligence - is the ability of Computers or machines to perform tasks that normally require human intelligence, such as learning, problem solving, understanding language and making decisions.

Types of AI

1. Narrow AI - is designed to perform one specific task or a limited set of tasks.
   - It cannot think or learn beyond its programmed purpose. e.g Siri
2. General AI (Strong AI)
   - General AI refers to Machines that can think, learn and understand anything a human can
   - It would have human like intelligence.
   example - Jarvis from Iron man.

Question 12.
What is Machine Learning.

Machine Learning - is a branch of Artifical Intelligence where computers learn from data and improve their performance without being explicitly programmed for every task.

How ML differ from traditional Programming

Traditional Programming	Machine Learning
write rules + instruction	Provide data + Expected outcome
Input data is processed using those rules.	The System learns patterns.
Output is produced	It creates its own rules to make predictions.

Three main types of Machine Learning

① Supervised Learning - model learns from labeled data - Input + Correct output are given.

② Unsupervised Learning - Model works with unlabeled data - It tries to find hidden patterns or groups.

③ Reinforcement Learning - The mode learns by trial and error using rewards and penalties.

Question 13.
What is deep learning.

Deep Learning is a type of technology that allows computers to learn from data and make decisions on their own, especially for complex tasks like reorganizing images, understanding speech or translating language.

How it relates to AI and Machine Learning

Artificial Intelligence - The big idea making machines behave intelligence.

Machine Learning is a part of AI where machine learn from data instead of being explicitly programmed

Deep Learning - A special type of machine learning that uses complex models to handle large and complicated data.

AI → ML → Deep Learning.

What are Neural Networks and How they work
- are the core technology behind deep learning.
- Computer System inspired by the human brain.

How they work.

- Input data is fed in, data passes through layers, weights and learning, output is produced and Improvement over time.

N.B *) Deep learning uses neural Networks to help machine learn complex patterns and make intelligence decision

Question 04.

What is Natural Language Processing.

NLP - is a field of computer science and AI that enables Computers to understand, interpret, and respond to human language.

Three real world applications.

(1) Voice assistance - Google Assistance uses NLP to understand your voice commands and responds.
(2) Autocorrect and Predictive text on Whatsapp
(3) Chatbots and messaging features Whatsapp with Smart reply.

(15)

Question 15.

What is Large Language Model LLM

LLM is a type of artificial intelligence that is trained on huge amounts of text data to understand and generate human language.

How is it different from Standard ML Model.

(1) Purpose -
ML model - designed for specific task.
LLM - Can handle many language tasks like answering questions, writing essay, translating and chatting.

(2) Data size.
ML model - Trained on smaller, task specific dataset
LLM - Trained on massive amount of data.

(3) Flexibility -
ML model - Needs retraining for new tasks.
LLM - Can perform different tasks without retraining.

Two well known LLMs and their uses.

(1) Chat GPT - used for answering questions, writing essays, Proofreading, coding help and others.
(2) Google Gemini - Search assistance, Content generation, and multimodal tasks.

Question 16.

What is Generative AI

- It is a type of artificial intelligence that can ~~retrieve~~ Create new content - such as text, images, music or code - based on patterns it has learned from existing data.

How does it work.

(1) Training on large data
(2) Learning patterns
(3) Generating Content.

Three tools that uses Generative AI

(1) Chat GPT
- Generates text based responses.
- Can Proofread documents.

(2) DALL.E
- Create images from text description

(3) Github Copilot
- Help developers write code.

Question 17

* What is training dataset.

- It is a collection of data used to teach an AI model how to recognize patterns and make decision

Why the quality of training data is so important

- Accurate Learning - Good data helps the model learn correct patterns.
- Better predictions: High quality data leads to more reliable results.
- Reduced error - Clean and well labled data prevents confusion.
- Generalization - work on new, unseen data.

What could go wrong if the data is biased or incorrect

(1) Bias in decision - If data favours one group over others, the AI will be unfair.
(2) Wrong Predictions - Incorrect data leads to inaccurate results.
(3). Poor performance in real life - If data doesn't represent real world situations, the AI may fail when used.
(4) Loss of trust - people stop trusting systems that make unfair or incorrect decision.

Question 18

What is Computer Vision.

- It is a field of AI that enables computers to see, understand and interpret image or video similar to how humans use their eyes and brains.

How Machines see and interpret images.

(1) Image capture - Camera captures an image and converts it into digital.
(2) Pixels and numbers - image is made up of tiny dots and pixels has numerical values.
(3) Processing with Algorithms - Computer uses techniques like ML and Deep learning to analyse patterns in those pixels.
(4) Feature detection - identifies edges, shapes, colors and textures.

Two real world

(1) Facial recognition - used by [redacted] phones or Samsung phones to open phones.

(2) Self driving cars - able to detect roads, pedestrians, robots and traffic signs.

Question 19

What is API (Application Programming Interface)

- It is a set of rules that allows different Software applications to communicate and share data with each other.

Simple real world analogy

- Ride apps like Uber or Bolt use map API to show locations and routes
- Paying for food using bank card at KFC connect APPs to to banks Securely.

Why API are important

* Connects different Systems - Allow apps and Servic to work together though built differently
* Save time and effort - Developers can use existing Services.
* Enable real time data sharing - Apps can fetch live data instantly. App on whether, Payment

Question 20

What is an API key?

- It is a unique code (like password) that identifies and authenticates an application when it requests data or services from another System.

Purpose of an API Key.

* Authentication - Confirms who is making the request
* Access control - determines what data or services the APP can use.
* Security - helps prevent unauthorized access.

How should API key should be Stored and protected

* Never hardcode them in public code. eg Github repositories.
* Store them in secure configuration files.
* Use Secret management tools. (cloud).
* Restrict usage (Limit by IP address, app).
* Rotate keys regularly (Change them periodically).

What could happen if exposed Publicly.

* Unauthorized usage - others can make request pretending to be your APP
* Unexpected costs - many APIs charge per request, so misuse can create large bills.
* Data breaches - Attackers may access Sensitive data.

Question 21.

REST API - is a way for different software System to communicate with each other over the internet using simple rules. It allows one applicant to request and send data to another.

What does REST Stands for.

REST Stands for Representational State Transfer.

Main HTTP methods used in REST API

① GET. - Purpose - Retrieve data
- use - When you want to fetch information
- Example - Get a list of users or view product.

② POST - Purpose - Create new data.
- Use - When you want to add something new.
- Example - Create a new user account.

(3) PUT - Purpose - updating existing data.
Use - When you want to modify existing information.
Example - Update user's profile.

(4) DELETE - Purpose - Remove data.
- use - When you want to delete something
- Example - Delete a user or remove a product.

Question 22

What is JSON (JavaScript Object Notation).
- It is lightweight format for storing and exchanging data. It represent data as key-value pairs, making it easy for both humans and computers to read and understand.

Why JSON is Commonly used

* Easy to read and write - Clear and Simple.
* Lightweight - use less data compared to other formats and its faster to send over internet.
* Language independent - It works with all programming languages.

* Perfect for API - when applications Communicate, Json is used to send and recieve data quickly and Clearly.

Simple example of Json object.

```
{
  "name" : "Khutso"
  "age" : 30,
  "is Student" : true,
  "Courses" : ["Data Science", "Data analytics"]

}
```

Question 23

Prompt engineering - is the skill of writing Clear, Spetific instructions to get the best possible respone from an AI tool like chatGPT or Claude.

Why Prompt engineering is important.

* Better results - Clears prompts lead to more accurate and helpful responses.
* Save time - you dont have to keep re-asking or collecting the AI.
* More Control - guide the tone, format and depth of an answer.
* Improves productivity - when Studying, coding, or writting good prompts make AI tools more powerful.

Two poor prompt.

1. Tell me about business - too vague - AI doesnt know which business.

(2). Write something about Sport - unclear and not Specific.

Two improved Versions.

(1) Explain how insurance business work Specifically re-insurance - question is Clear.

(2). Explain why Soccer in EPL is better than in South Africa.

Question 24.

MAP function definition

- It is a programming concept used to apply a specific operation to every item in a list and return a new list with results.

What it does in programming

- process each element one by one.
- Applies a function to each element
- Returns a new Collection with the updated values.
- Do not Change the original data.

∴ Python example.

```
data = [ "alice", "Bob", "Charlie" ]
Print(data)
```

Javascript

```
Let data = ["Alice", "Bob", "Charlie"];
Console.log(data);
```

Question 25:

What is cloud computing:

- It is the delivery of computing services (like storage servers, software, and databases) over the internet instead of using your own physical computer.

Three main cloud Service Models

(1) Iaas (Infrastructure as a Service) - provides basic computing resources like virtual machines, storage and networks. eg Amazon Web Service (AWS) EC2.

(2) Paas (Platform as a Services) - Provides a platform for developers to build, test and deploy applications without worrying about infrastructure. eg Google App.

(3) Saas (Software as a Service) - This delivers ready to use software over the internet.
eg email

Question 26.

What is Version Control System?

- It is a tool that helps you track changes in files over time. It allows you to see what was changed, who changed it and even go back to earlier version if needed.

What is Git

- It is a popular version control System that helps people manage and track changes in their code or projects

Why used by data analysts and developers

- Tracks changes easily - Can see edit made.
- Collaboration - multiple people can work on the same projects without conflicts.
- Backup and Safety - work saved and can restore older version.

*Works with platforms like Github for sharing and teamwork.

What is Commit.
- is saved version of your work.

What is Branch.
- It is a separate copy of your project where you can make changes without affecting the main version.

Question 27.

What is Jupiter Notebook

- A notebook that is interactive tool that let you write code, run it, and see results - all in the same document.
- Easy to combine codes results and Notes.

~~Mostly~~

What programming is commonly used with.
- Jupiter Notebook is most commonly used with Python although other language like R and Julia can use it.

Why its popular among data analysts and data scientist.

- Interactive and easy to use - Can run code step by step and immediately see output.

- Combines code and explanation - can write notes along side the code which helps understand the work
- Great for visualization - Charts and graphs can be displayed directly in the Notebook.
- Supports data science libraries - works well with powerful python tools like Pandas, Numpy and Matplotlib.
- Easy to Share - Notebooks can be shared with others, making collaboration simple.

Question 28.

What is python.

- Python is high level programming language that is easy to read and write.
  - You can use python to build websites, automate tasks, analyse data and create artificial intelligence systems.

Why python is popular for data analytics and AI

- Easy to learn and use - Its simple syntax looks similar to plain english.
- Powerful libraries - Python has many ready made tools that make data analysis and AI faster and easy.
- Large community support - millions of developers use python.
- Strong industry adoption - Companies like Google, and Netflix use python for data driven task

Four Common Libraries:

(1) Numpy - used for working with number and arrays. Help perform fast mathematical Calculations
(2). Pandas - Used for handling and analysing data in tables. Great for Cleaning and organizing data.
(3) Matplotlib - Used to create charts and graphs to visualize data.
(4) Scikit-learn - used for building machine learning like predictions and Classifications.

Question 29. SECTION D

What is data privacy?

It is about protecting a person's personal information and making sure it is collected, used and shared properly and safely.

Why data Privacy is important

- Protects personal information - prevents misuse of sensitive data like bank details.
- Prevents identity theft and fraud - keeps criminals from stealing and abusing personal data.
- Builds trust: People trust companies more when their data is handled responsibly.
- Legal Compliance - Companies must follow laws or face penalties.

What is GDPR

The General Data Protection Regulation (GDPR) is a data protection law created by the European Union (EU) that came into effect in 2018.

Key features of GDPR

- Consent is required, right access, right to be forgotten, data protection responsibility, and heavy fines for violations

Which Countries does GDPR

- France, Italy, UK, South Africa, Germany and Hungary.

Question 30.

What is data Ethics

- It refers to the principles and standards that guide how data is collected, used, shared, and protected in a way that is fair, responsible, and respectful of people's rights and privacy.

Three examples of unethical use of data or AI

(1) Misusing personal data without consent

What happened - personal data used for political advertising without proper consent.

Why its a problem - violates privacy, Can manipulate opinions and behavior

(2). Biased AI decision-making.

what happend - An AI hiring system favoured male candidates because it was trained on biased historical data.

Why its problematic - leads to unfair discrimination, reinforces existing inequalities and Prevent equal opportunities.

(3) Surveillance and tracking without transparency.
What happened - people are monitored in public or online spaces without knowing how their data is used
Why its problematic - Invades privacy, can be used for control or misused of power and lack of accountability and transparency.

Question 31

What is data governance.
- It is the set of rules, processing and responsibilities that ensure an organization's data is accurate, secure, consistent and used properly.

Key components of a good data governance framework.

① Data quality.
- Ensures data is accurate, complete, consistent and up-to-date.
(2) Data policies and standards.
- Rules for how data should be collected, stored, used and shared.
(3) Data Security and privacy
- Access control, encryption and compliance with laws like GDPR
(4) Roles and Responsibilities -
- defines who is responsible for data, data owners, data stewards and users
5 Data Management processes.
processes for storing, updating, backing up data

Question 32.

What is Bias in AI

- It refers to a situation where an AI system produces unfair, skewed or prejudiced results, often favouring or disadvantaging certain groups of people.

What it means for an AI model to be biased

- considered biased when:
* It makes systematically unfair decisions.
* Its results are influenced by imbalanced or flawed training data.
* It reflects human prejudices present in the data it learned from.

Real-world example of AI Bias.

* Amazon developing an AI system to screen job applications.
  - System trained on past data.
  - Historical hire were men.

Potential Consequences of AI Bias.

* Unfair treatment
- Certain groups may be unfairly rejected
* Discrimination
- Bias can reinforce gender, social inequality
* Loss of trust
- people may stop trusting AI systems if they are seen as unfair.
* Legal and ethical issues.
- Companies can face lawsuits.

Question 33.

Maine differences between data analyst data Scientist, data engineer and AI engineer

Roles	main Skills	What the do	Common tools
Data analyst	Excel, SQL, basic stats, data visualization	Analyzes data to answer questions and create reports	Excel, SQL, Tableau and Power BI
Data Scientist	Python/R, Statistics, Machine learning.	Builds models to predict the future collects store and Process outcomes and find patterns.	Python, Tensorflow Jupyter Notebook
Data engineer	Databases, ETL pipelines, Big data tools	Builds System to collect, store and process data	Apache Spark, Hadoop, SQL
AI engineer	Machine learning, deep learning, Software Engineering	Develops AI systems and integrates models into applications	Pytorch, Tensorflow, Python.

Question 34.

What is business intelligence.

It is the process of collecting, analyzing and presenting data to help businesses make better decisions.

Two popular

(1) Microsoft Power BI
- Turns data into interactive dashboards and reports.
- Connects for many data sources (Excel, databases, Cloud Services).
- Help users visualize data using charts and graphs.

(2) Tableau
- Known for powerful and easy-to-use data visualization
- Allows users to explore data with drag and drop features.

How they help Companies

- Company can visualize which regions have the highest sales using maps and charts.
- Managers can analyse customer behavior and spot trends.

SECTION E

Question 35

* Learning the difference between data science, data analytics, data engineerg and AI engineer because these are very related and a dedicated individual can become either one.
* Python is must learn for data science and SQL is in most professions.
- ~~Would love to~~
- Loved learning data engineering and how it works. Courses are interelated!

Question 36.

+ Javascript example was quite difficult to understand.
- used ChatGPT to understand the concept even though it is still difficult to grasp

Question 37

* I'm current a Senior investment analyst and would love to move to quants hence im studying data science. Even if I start at data science but the ultimate goal is to move into quants and become quants analyst in asset management.

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